Alessandro De Bellis

PHD RESEARCHER IN NLP AND LANGUAGE MODELS

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PROFILE

I am a PhD researcher focused on AI research in pretrained language models, interpretability, probing, and knowledge graph reasoning. Recent work examines how frozen PLM representations encode semantics, factuality, and affective signals, with applications in retrieval and recommendation.

Available for work: Looking for Postdoctoral and industry research roles, remote or relocating.

EDUCATION

2022 – 2026 **PhD, Industry 4.0 (Information Processing Systems)**
Politecnico di Bari, Department of Electrical and Information Engineering
Thesis: A Journey through the Hidden Representations of Pretrained Language Models: Semantics, Factuality and Beyond. Tutors: Prof. Tommaso Di Noia and Prof. Eugenio Di Sciascio.

2018 – 2022 **MSc, Computer Science Engineering**
Politecnico di Bari, Department of Electrical and Information Engineering
Thesis: Interpreting the Latent Semantics of Transformers: A Focus on BERT Explained through Freebase. Tutor: Prof. Tommaso Di Noia

2014 – 2018 **BSc, Computer Science Engineering**
Politecnico di Bari, Department of Electrical and Information Engineering

EXPERIENCE

2022 – 2025 **Doctoral researcher**
Department of Electrical and Information Engineering, Politecnico di Bari
Research on probing the frozen hidden representations of language models: semantics and knowledge graphs, factuality, sentiment and emotion, and zero-shot recommendation and search.

2019 **Software engineer (intern)**
Finlogic Spa, Acquaviva delle Fonti

2018 – 2019 **Full-stack web developer**
Icommlab Srl, Acquaviva delle Fonti

TEACHING

2024/2025 **Supplementary teaching activities: Algorithms and Data Structures in Java**
Politecnico di Bari · BSc in Computer and Automation Engineering [LT17] (D.M. 270/04)
40 hours.

2023/2024 **Supplementary teaching activities: Databases and Information Systems**
Politecnico di Bari · BSc in Computer and Automation Engineering [LT17] (D.M. 270/04)
40 hours.

PEER REVIEW

2026 **EMNLP, NeSy, CIKM, The Web Conference (WWW)**

2025 **ACM RecSys, ECIR**

2024 **EKAW, ACM SIGIR**

PUBLICATIONS (11)

[11] 2026 CONFERENCE

Do Composed Image Retrieval Benchmarks Require Multimodal Composition?
Matteo Attimonelli, **Alessandro De Bellis,** Aryo Pradipta Gema, Rohit Saxena, Monica Sekoyan, Wai-Chung Kwan, Claudio Pomo, Alessandro Suglia, Dietmar Jannach, Tommaso Di Noia, Pasquale Minervini
NeurIPS 2026 · Accepted arxiv.org/abs/2605.14787
MY CONTRIBUTION Conceptualization, Data curation, Methodology, Validation, Writing - original draft, Writing - review & editing

[10] 2026 CONFERENCE

PUMA: Post-Hoc Sparsification of Universal Multimodal Embeddings for Efficient Retrieval
Matteo Attimonelli, **Alessandro De Bellis,** Franco Maria Nardini, Claudio Pomo, Cosimo Rulli, Rossano Venturini, Tommaso Di Noia
Findings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP) · Accepted
openreview.net/forum?id=BkDE7OPg7m
MY CONTRIBUTION Validation, Writing - review & editing

[9] 2026
PHD THESIS

A Journey through the Hidden Representations of Pretrained Language Models: Semantics, Factuality and Beyond

Alessandro De Bellis

PhD thesis, Interuniversity Ph.D. Program in Industry 4.0, Politecnico di Bari hdl.handle.net/11589/295622

[8] 2025
CONFERENCE

Type-Less yet Type-Aware Inductive Link Prediction with Pretrained Language Models

Alessandro De Bellis, Salvatore Bufi, Giovanni Servedio, Vito Walter Anelli, Tommaso Di Noia, Eugenio Di Sciascio

Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), Suzhou, China doi:10.18653/v1/2025.emnlp-main.1383 aclanthology.org/2025.emnlp-main.1383/

MY CONTRIBUTION Conceptualization, Formal analysis, Methodology, Investigation, Software, Validation, Visualization, Writing - original draft, Writing - review & editing

[7] 2025
CONFERENCE

Do We Really Need Specialization? Evaluating Generalist Text Embeddings for Zero-Shot Recommendation and Search

Matteo Attimonelli, **Alessandro De Bellis,** Claudio Pomo, Dietmar Jannach, Eugenio Di Sciascio, Tommaso Di Noia

Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25) doi:10.1145/3705328.3748040

MY CONTRIBUTION Conceptualization, Methodology, Software, Validation, Visualization, Writing - original draft, Writing - review & editing

[6] 2025
CONFERENCE

Are the Hidden States Hiding Something? Testing the Limits of Factuality-Encoding Capabilities in LLMs

Giovanni Servedio, **Alessandro De Bellis,** Dario Di Palma, Vito Walter Anelli, Tommaso Di Noia

Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025), Vienna, Austria doi:10.18653/v1/2025.acl-long.304 aclanthology.org/2025.acl-long.304/

MY CONTRIBUTION Conceptualization, Data curation, Formal analysis, Methodology, Investigation, Software, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing

[5] 2025
CONFERENCE

LLaMAs Have Feelings Too: Unveiling Sentiment and Emotion Representations in LLaMA Models Through Probing

Dario Di Palma, **Alessandro De Bellis,** Giovanni Servedio, Vito Walter Anelli, Fedelucio Narducci, Tommaso Di Noia

Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025), Vienna, Austria doi:10.18653/v1/2025.acl-long.306 aclanthology.org/2025.acl-long.306/

MY CONTRIBUTION Conceptualization, Validation, Writing - review & editing

[4] 2024
CONFERENCE

PRONTO: Prompt-Based Detection of Semantic Containment Patterns in MLMs

Alessandro De Bellis, Vito Walter Anelli, Tommaso Di Noia, Eugenio Di Sciascio

The Semantic Web – ISWC 2024: 23rd International Semantic Web Conference, Baltimore, MD, USA, November 11–15, 2024, Proceedings, Part II Pages 227 - 246 doi:10.1007/978-3-031-77850-6_13

MY CONTRIBUTION Conceptualization, Formal analysis, Methodology, Investigation, Software, Validation, Visualization, Writing - original draft, Writing - review & editing

[3] 2023
CONFERENCE

Semantic Interpretation of BERT Embeddings with Knowledge Graphs

Alessandro De Bellis, Giovanni Maria Biancofiore, Vito Walter Anelli, Fedelucio Narducci, Tommaso Di Noia, Azzurra Ragone, Eugenio Di Sciascio

Proceedings of the 31st Symposium of Advanced Database Systems (SEBD 2023), CEUR Workshop Proceedings, vol. 3478 PDF

[2] 2023
CONFERENCE

Structuring the Unstructured: an LLM-guided Transition

Alessandro De Bellis

Proceedings of the Doctoral Consortium at ISWC 2023, CEUR Workshop Proceedings, vol. 3678 PDF

[1] 2022
CONFERENCE

Interpretability of BERT Latent Space through Knowledge Graphs

Vito Walter Anelli, Giovanni Maria Biancofiore, **Alessandro De Bellis,** Tommaso Di Noia, Eugenio Di Sciascio (authors listed in alphabetical order)

CIKM '22: Proceedings of the 31st ACM International Conference on Information & Knowledge Management, Pages 3806 - 3810 PDF

MY CONTRIBUTION Conceptualization, Methodology, Investigation, Software, Validation, Visualization, Writing - original draft, Writing - review & editing

SKILLS

Languages	Italian (native), English (fluent)
ML & research	Python, PyTorch, Hugging Face Transformers, scikit-learn, NumPy, LaTeX
Tools	Git, Linux, SLURM, Docker
Topics	LLM interpretability, Representation probing, Knowledge graph reasoning, Inductive link prediction, Zero-shot retrieval and recommendation, Vision-language models, Evaluation
Web	HTML, CSS, PHP, JavaScript, Apache Web Server